higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1: TIMBER COSTRUCTION

1.1	1.1.1	I		
	1.1.2	C		
	1.1.3	E		
	1.1.4	B		
	1.1.5	G		
	1.1.6	J		
	1.1.7	H		
	1.1.8	F		
	1.1.9	A		
	1.1.10	D		
			(10 × 1)	(10)

1.2	1.2.1	False		
	1.2.2	False		
	1.2.3	False		
	1.2.4	False		
	1.2.5	False		
			(5 × 1)	(5)
				[15]

QUESTION 2: METALS

2.1
- Keep your cutting tools in good condition.
- Receive proper training before operating cutting tool.
- Beware of sharp edges, crush points and pinch points when using cutting machine.
- Do not leave cutting tools unattended.
- Always wear safety protective equipment. (5)

2.2 Pure metal is a metal that is found in nature and does not intentionally contain any other materials✓✓ while alloy is made up of two different chemical elements of which one is a metal.✓✓ (4)

2.3 2.3.1 Drawing: a process of changing metal properties by means of tensile force.✓ Material is drawn through die, which reduces the cross-section area of wire, a bar or tube without any pre-heating.✓it is usually done at room temperature but can also done at elevated temperature to reduce the tensile forces required to stretch the material.✓

2.3.2 Annealing: the process of changing metal properties by heat treatment.✓ The method changes properties such as strength and hardness.✓ It softens the metal to prepare it for shaping, stamping or forming.✓

2.3.3 Work-hardening: This method changes metal properties by straining it beyond its yield point.✓ It can be done on purpose to strengthen a metal but can also be done accidentally.✓ By exerting stress, plastic deformation occurs and the metal becomes stronger and difficult to deform.✓

(3 × 3) (9)

2.4 Punching is the process by which a metal sheet is perforated. (1 × 2) (2)
[20]

QUESTION 3: STEEL CONSTRUCTION

3.1 3.1.1 Elastic behavior: where steel changes its shape temporarily due to stress applied and goes back to its original shape once the force is removed.

3.1.2 Toughness: when steel is able to resist braking when it is stressed.

3.1.3 Yield point: where the steel reached maximum strength and started to deform plastically, which means it will never return to its original shape.

3.1.4 Young's modulus: measurement of the stress per unit of strain. It indicates the resistance of the steel to deformation within its elastic limit.

3.1.5 Elongation: refers to how much steel can stretch before it reaches the point where it finally breaks.

(5 × 2) (10)

3.2
- Save time and money.
- Minimise on-site construction and overcrowding on the construction site.
- Cut down on the number of laborers needed on site.
- Minimise the need for formwork, shuttering and scaffolding on the site.
- Benefit from the availability of skilled labour in factories.
- Reduce the time spent in bed weather. (Any 5 × 1) (5)

3.3
- It is light.
- No formwork is needed.
- It is not flammable.
- It can be recycled.
- It is of a uniform quality.
- There is little delay due to bad weather. (Any 5 × 1) (5)

[20]

QUESTION 4: UNDERSTANDING AND COMBATING CORROSION OF METALS

4.1
- All work has been completed as stipulated in the contractual agreement.
- The area has been left in a safe condition.
- Any defective work will be fixed.
- All hazards or potential hazards arising from the work have been controlled.
- Bolts inserted and tightened.
- Steelwork levelled and aligned. (Any 5 × 1) (5)

.4.2 A braced bay is a bay that has been temporarily or permanently strengthened by a piece of structural material. (2)

4.3
- Support the structure.
- Prevent buckling of web members loaded in compression.
- Share loads between adjacent members.
- Transfer lateral forces to diaphragms.
- Restrain overall lateral displacement.
- Enhance the basic soundness of the entire structural system. (Any 5 × 1) (5)

4.4
- Mill rolling tolerances
- Fabrication tolerances
- Erection tolerances (3)

[15]

QUESTION 5: JOINING METALS BY SOLDERING, BRAZING AND WELDING

5.1 5.1.1 Welding: is the process of joining two or more metals together by melting a third material called filler.✓ The filler forms a molten weld pool that is then act as a glue to join metals.✓

5.1.2 Brazing: is the process of joining two metals together by melting third material called filler.✓ It is done at a very high temperature with map gas torch and the bond formed is much stronger. ✓

5.1.3 Soldering: is the process of joining two metals together by melting third material called filler. It is done using soldering iron✓ at a lower temperature.✓

(3 × 2) (6)

5.2
- Welding helmet.
- Safety goggles.
- Leather apron.
- Safety boots.
- Heat resistant cotton clothing.
- Leather welding gloves. (Any 5 × 1) (5)

5.3 5.3.1 Shielded metal arc welding.

5.3.2 Gas metal arc welding.

5.3.3 Flux cored arc welding.

5.3.4 Tungsten inert welding.

5.3.5 Submerged arc welding.

(5 × 1) (5)

5.4
- E = Electrode.
- 60 = Minimum tensile strength in thousands of pounds.
- 1= Recommended welding position.
- 13 = Notes. (4 × 1) (4)

[20]

QUESTION 6 PROPETRIES OF GLASS AND THE CUTTING, DRILLING AND SHAPING OF GLASS

6.1 6.1.1 Toughened glass. > shower doors,office partitioning and large windows.

6.1.2 Laminated glass > glass doors, shower doors, shop front and car windscreens.

6.1.3 Wired glass > glass doors and in roofing.

6.1.4 Glazed panels > mounted in wood and construction of walls

6.1.5 Float glass > domestic housing windows and display glass

(5 × 1) (5)

6.2
- It is resistant to explosions.
- It blocks unwanted noise.
- It prevents unlawful entry.
- It is resistant to windstorms.
- Its thicker type resists impact from bullets.
- The layer between the layers of glass acts as a barrier. (Any 5 × 1) (5)

[10]

TOTAL: 100